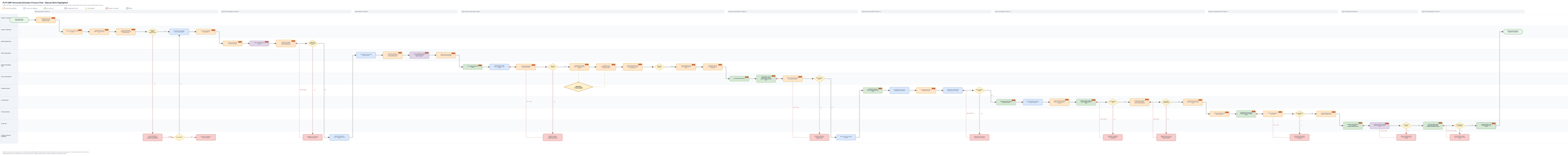

PLPI GMP Horizontal Swimlane Process Flow - Manual Work Highlighted

Based on SOP stages 1-14 and handwritten operational notes. Orange boxes indicate manual/operator or paper-based work; blue boxes indicate PI system/scanning; green boxes indicate QA/GMP controls; red boxes indicate exception/deviation/rework paths.

🟧 Manual work / paper handling　🟦 PI system / scan / digital status　🟩 QA / GMP control　🟪 Controlled document / record　◇ Decision gateway　🟥 Exception / hold / deviation　🟩 Release

Stage 1 Goods InSOP_PLPI_0002_v34
Stage 2 Batch CheckingSOP_PLPI_0005_v22
Stage 3 BARSOP_PLPI_0006_v16
Stages 4-8 Printing / Cartons / Braille / Folding
Stage 9 Pre-Assembly QCSOP_PLPI_0052_v22
Stage 10 Production ControlSOP_PLPI_0037_v14
Stage 11 AssemblySOP_PLPI_0007_v25
Stage 12 Post-Assembly QCSOP_PLPI_0053_v15
Stage 13 Pre-QP QASOP_PLPI_0009_v24
Stage 14 QP CertificationSOP_PLPI_0010_v13
Supplier / Transport
Goods In / Warehouse
Batch Checking Team
BAR / Documentation
Printing & Packaging Prep
QA / Pre-Assembly QC
Production Control
Assembly Room
Post-Assembly QC
Pre-QP / QP
PI System / Records / Exceptions
Start: supplier / third-party stock received
MANUAL
Deliver stock with invoice, packing list, PO and temperature evidence
Receive goods and open Goods In process
Verify delivery note, PO, supplier, invoice and packing list
Check quantity, damage, temperature, cold-chain / CD and FMD indicators
Discrepancy / damage / temperature issue?
No
Yes
HOLD / QUARANTINE Log deviation or temperature excursion; notify QA/Planner
Investigate
Issue resolved?
No
Reject / return / destruction / supplier investigation
Yes
Record Goods In, apply pallet / box ID and barcode labels
Move stock to quarantine / holding location
Retrieve one batch with documents and sample
Print / review Product Check Log (PCL)
Verify product, strength, batch, expiry, EC/MA, PL, country and leaflet details
Batch details match PCL / pack / system?
No
resolved / recheck
Regulatory query / batch hold / quarantine until approved
No
Add / confirm batch in PI system and move to Waiting for BAR
Create BAR in PI system / B&S Batch Add module
Verify PCL, sample pack, invoice, temperature records and required documents
Print controlled BAR, approved cartons / peel-out / braille pages as required
Place BAR, PCL, invoice and sample in closed zipped wallet
Line clearance before label printing
Scan BAR / batch in printer module and display required labels
Verify label details against BAR and sample pack
Label details correct?
Yes
No
reprint / recheck
Printing error: log issue, quarantine components, regulatory request / reprint
Print labels; verify first/last labels, record actual print and spoilage
Print leaflets; verify reference, size, product, PL and mechanical index
Issue pre-printed cartons; scan BAR; verify reference, quantity and sample carton
Relabelling or Repackaging?
Braille label required?
Yes
Print / emboss braille; verify against Braille Declaration Sheet
Fold leaflets; verify first fold, orientation, reference and quantity
Pre-assembly line clearance
Verify BAR, PCL, invoice, labels, leaflets, cartons, braille, mock-up and sample pack
Prepare / confirm specimen pack and regulatory evidence
Pre-Assembly QC passed?
No
correct / release
QA hold: missing / incorrect components, reissue or regulatory review
Yes
Mark batch Ready for Assembly in PI system
Production Control line clearance and one-batch-only check
Use handheld Stock Take Out; scan BAR, box ID and location
Verify product, batch, IMP no., quantity and location
Transfer stock to Assembly Area and allocate room in PI system
Room / sequence conflict?
No
Yes
approved sequence
Manager approval / sequence change; log reason for priority
Assembly room line clearance and start-up checks
Scan BAR and operator passcode; start batch in PI system
Assemble product; apply labels, leaflet, carton, seals and pack components
Perform timed IPC checks and capture required evidence / photo
IPC completed on time?
Yes
No
unlock / repeat IPC
System lock / investigation / authorised unlock before continuation
Complete batch reconciliation; product, labels, leaflets, cartons, surplus and damage
Reconciliation within tolerance?
Yes
No
resolve variance
Investigate variance; document damage / missing / residue; deviation if needed
End batch in system; review BAR; send batch and documents to QA
Receive assembled stock and batch documentation
Verify packs, boxes, specimen, reconciliation and AVC sample checks
Print and verify Quarantine Stock labels
Post-Assembly QC passed?
Yes
No
correct / rework
QA hold / rework / quarantine incomplete packs and correct documentation
Place completed stock in QA quarantine / awaiting Pre-QP
Pre-QP QA review: BAR, PCL, IPC, reconciliation, deviations, retention sample
Generate QP Release Log / Batch Summary and compile release pack
Pre-QP review passed?
Yes
No
corrections complete
Return to responsible area for correction / deviation closure / evidence
QP review: supplier status, supply chain, licence, GMP evidence, CAPA/deviation status
QP satisfied batch is compliant?
Yes
No
evidence / rework complete
QP reject / rework / request further evidence; batch remains on hold
QP certifies batch; record release log; release in PI system
Released goods transferred to warehouse / distribution

Manual-work hotspots: Goods In inspection, PCL checks, BAR wallet preparation, label/leaflet/carton verification, braille verification, leaflet folding, Pre-Assembly QC, assembly operations, IPC, reconciliation, Post-QC, Pre-QP and QP review.

Recommended improvement focus: validated EBR, scan-to-proceed gates, electronic reconciliation, digital line clearance, QP readiness dashboard and deviation/CAPA workflow.